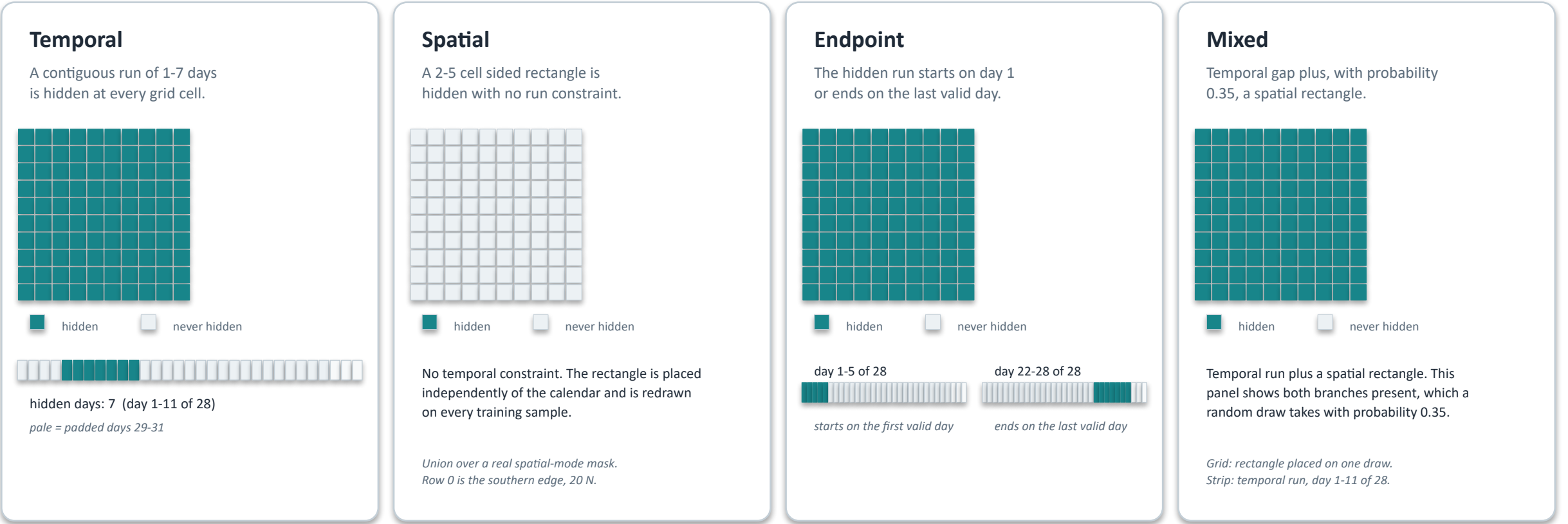


Missing figure 1 — Artificial gap regimes used in the evaluation

Daily NASA POWER radiation on a 10 x 10 grid, 20-30 N / 100-110 E. Every evaluated gap is synthetic; the manuscript defines four families but currently shows none of them.



Protocol. Masks are drawn once per month by random_observation_mask (dmgsr.masked_temporal) with gap lengths 1-7 days, spatial side lengths 2-5 cells, and spatial probability 0.35 in mixed mode. The same stored mask is scored for every method, and only valid, artificially hidden cells enter RMSE, CRPS, coverage and interval width. Patterns shown here. Each calendar strip is one regenerated example of that family over a 28-day month; the spatial panel shows the union of a real spatial-mode mask. The Mixed panel can therefore coincide with the Temporal strip for a draw in which the spatial branch was not taken.

Boundary. This figure depicts the experimental design only and reports no skill. Aggregate skill is in Table 1 and Figures 3-6.

Missing figure 2 — What happens at a chunk boundary during reverse sampling

64 cosine-scheduled reverse steps are processed in eight chunks of eight. After each chunk the sequence below runs once, then a consistent noisy latent is rebuilt and fed into the next chunk.

A. Within a chunk - eight denoising steps

velocity prediction v_{θ} -> clean estimate $r_0 = \sqrt{\alpha_t} z - \sqrt{1 - \alpha_t} v_{\theta}$ -> scale back to radiation and clamp to $[0, U]$, $U = \max(50, 1.05 \max(R_a, 20))$ -> restore observed cells -> recompute the noise direction and re-noise the latent

B. At the chunk boundary - correction sequence, in code order

1. Ensemble spread

σ^2 = variance over the 12 members of the current clean estimate

2. Variance-adaptive gain

$k = \text{clip}(\sigma^2 / (\sigma^2 + (s_r f)^2), 0.15, 0.85)$
 $\text{progress} = 1 - \text{low} / 64$

3. Bounded monthly projection

allocate the coarse block energy remaining after the observed contribution, using the actual valid-day count, then clamp to U

4. Graph boundary update

move hidden boundary cells toward the four-neighbour context, strength 0.15, then restore observations

5. Rebuild a consistent latent

recompute eps from the corrected state, re-noise at α_{low} , re-noise observations, feed into the next chunk

C. After step 0 - final processing

bounded monthly projection (applied twice, with the spatial blends between them)
spatial boundary blend
spatial diffusion blend, 4 iterations
restore observed cells and clamp to $[0, U]$

Boundary of this description

Operations 2-4 are corrections layered on implicit diffusion sampling; they are not part of the learned denoiser and they alter the output distribution.
Finite alternating projections do not establish exact simultaneous satisfaction of every constraint, and this sequence is not a validated graph ablation.

Graph regularization ablation — four matched configurations

Every factor except the graph mechanism is held fixed: the same data, chronological split, mask schedule, fitting seeds, epoch count and checkpoint-selection rule. The graph mechanism has a training part and a sampling part, so each is removed separately.

Configuration	Structural penalty (training loss)	Graph boundary + spatial blends (sampling)	What it isolates
A Full	0.05	strength 0.15, blends 0.25	the evaluated model
B No sampling graph	0.05	removed	inference-time graph correction
C No training penalty	removed	strength 0.15, blends 0.25	the structural loss term
D No graph at all	removed	removed	both mechanisms together

Held fixed

POWER daily radiation, 20-30 N / 100-110 E
fitting 2000-2009, selection 2010, reporting 2011
3 fitting seeds (11, 22, 33), 30 epochs each
sampling: 12 members, 8-step chunks, 64 steps
training curriculum: spatial probability 0.8, missing-cell weight 5.0, gradient weight 0.08

What it does not establish

2011 was inspected during development, so this is a development comparison.
One stored mask per month; the bootstrap resamples months, not independent gaps.
The graph is the fixed four-neighbour grid: no learned attention, no irregular network.
Removing a penalty also changes the effective loss scale; coefficients were not retuned.

Results, 2011 reporting cells (547)

Configuration	RMSE	CRPS	Cov. 90%	Edge MAE	Lap. MAE
A Full	29.76	17.56	79.1%	20.04	50.67
B No sampling graph	29.83	17.60	79.3%	20.12	50.84
C No training penalty	29.75	17.55	78.8%	19.98	50.41
D No graph at all	29.82	17.59	79.0%	20.06	50.58

Grid-graph structure on the same cells

Verdict. The four configurations are indistinguishable: RMSE spans 29.75-29.83 W/m2 and CRPS 17.56-17.60, while the between-seed standard deviation inside a single configuration is 1.4 W/m2 in RMSE. Every paired month-block bootstrap interval for full minus ablated touches or straddles zero.

Provenance and placement for the added figures

These figures are new. They are not part of the current eight-figure manuscript and are not yet cited in it.

Figure	Subject	Intended placement	Source of every value shown	What it does not claim
G1	Missingness regimes	Data and evaluation design, after the mask paragraph	random_observation_mask in dmgsr.masked_temporal; parameters 1-7 days, 2-5 cells, 0.35 spatial probability	Design only; reports no skill
G2	Chunk-boundary correction loop	Methods, Graph penalties and constraint feedback	sample_masked in dmgsr.masked_temporal; chunk size 8, gain clip 0.15-0.85, graph strength 0.15	Description only; no ablation
G3	Graph-regularization ablation	Results, a new subsection after the matched-score results	dmgsr.ablation; four matched training runs, validation CRPS selection on 2010, scoring on 2011	Development-period comparison only

Omissions that no drawing can close

- 1. Independent validation. Calibration uses 2010 and reporting uses 2011; both were inspected during development, so G3 and the main table are development comparisons. A confirmatory figure needs the frozen pipeline re-run on a new region or period, ideally against an independently defined missingness process.
- 2. Sensitivity to the monthly condition. The monthly field is built from the complete reference product and therefore carries information about the hidden values. Quantifying that dependence needs new inference with degraded or partially observed aggregates.
- 3. External comparators. No independently implemented probabilistic imputer (for example CSDI) was trained on this protocol, so no cross-method claim is made.

Cross-check of the present results. 46 numeric claims in Manuscript.tex (the main table, the deltas, the bootstrap interval, the selected calibration parameters and the physical-support count) were re-derived from paper/evidence, and agree within the reported precision; a further 7 trace and fixed-case quantities in Figures 7-8 were reproduced the same way. The audit scripts are paper/audit_numbers.py, paper/audit_figures.py and paper/audit_monthly.py.